

Changing Earth's Surface

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS **Weathering** breaks rock into smaller pieces. **Erosion** carries the pieces away by wind, water, or ice. **Deposition** drops the pieces in a new place, building things like sandbars and deltas. Changes can be **slow** (a canyon carved over ages) or **fast** (an earthquake or a volcano).

Round 1 — Retrieve It

1. What is it called when rock is broken into smaller pieces?

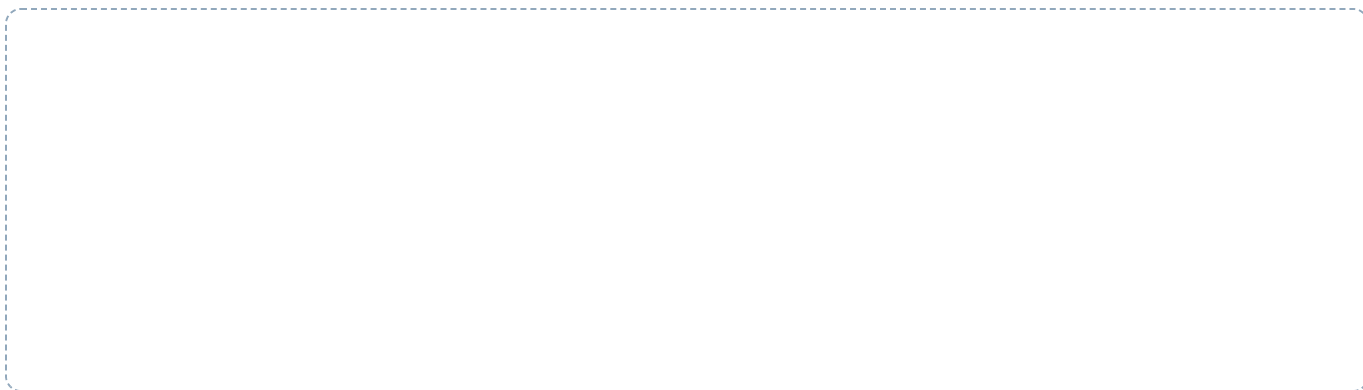
2. What carries the broken pieces away to a new place?

3. Name one thing that can cause erosion.

4. Give one example of a fast change to Earth's surface.

Round 2 — Sketch It

Draw a river carrying sand and dropping it to build a delta. Use arrows to show erosion (carrying) and label where deposition (dropping) happens.



Changing Earth's Surface — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. Wind blows sand against a rock for many years until the rock crumbles. Which process is this, and what happens to the rock?

2. A river drops sand at its mouth and slowly builds a delta. Which process builds the delta, and how?

3. A volcano erupts and covers a valley with lava in one day. Is this a slow change or a fast change? Explain.

Round 4 — Reason from Evidence

The Grand Canyon is very deep, but a river runs through it. How could a river make such a huge canyon? Use weathering and erosion to explain, and tell whether this was slow or fast.

Changing Earth's Surface — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain the order of weathering, erosion, and deposition. Tell what happens in each step.

C**Connect**

Connect erosion and deposition. If pieces are carried away from one place, what happens to them somewhere else?

E**Extend**

Some changes are slow and some are fast. Sort these into slow or fast: an earthquake, a canyon forming, a volcano, a beach worn down over years. Explain your choices.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Changing Earth's Surface

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Weathering.
2. Erosion.
3. Any: wind, water, or ice (glaciers).
4. Any: an earthquake or a volcano (eruption).

Sketch It — look for

A river with arrows showing pieces being carried (erosion) and a delta drawn where the river drops the sand (deposition), with deposition labeled correctly.

Apply It

1. Weathering; the rock is broken into smaller pieces.
2. Deposition; the river drops (deposits) the sand it was carrying, and it piles up to form the delta.
3. A fast change; it changed the land in a very short time (one day).

Reason from Evidence — look for

The river slowly weathered and eroded the rock, carrying pieces away bit by bit over a very long time (millions of years). This is a slow change.

ACE — Extend look-for

Fast: earthquake, volcano. Slow: canyon forming, a beach worn down over years. Reasoning based on how quickly each change happens.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.